OpenLearn | Free learning from The Open University

The Open University

Statement of participation

Mahdiyeh Ghasemi

has completed the free course including any mandatory tests for:

Exploring data: Graphs and numerical summaries

This 20-hour free course introduced a number of ways of representing data graphically and summarising data numerically.

Issue date: 29 September 2026



www.open.edu/openlearn

This statement does not imply the award of credit points nor the conferment of a University Qualification.
This statement confirms that this free course and all mandatory tests were passed by the learner.

Please go to the course on OpenLearn for full details:
https://www.open.edu/openlearn/science-maths-technology/mathematics-statistics/exploring-data-graphs-and-numerical-summaries/content-section-0

COURSE CODE: **M248_1**

OpenLearn | Free learning from The Open University

The Open University

Exploring data: Graphs and numerical summaries

https://www.open.edu/openlearn/science-maths-technology/mathematics-statistics/exploring-data-graphs-and-numerical-summaries/content-section-0

Course summary

This free course, Exploring data: graphs and numerical summaries, will introduce you to a number of ways of representing data graphically and of summarising data numerically. You will learn the uses for pie charts, bar charts, histograms and scatterplots. You will also be introduced to various ways of summarising data and methods for assessing location and dispersion.

Learning outcomes

By completing this course, the learner should be able to:

- understand and use standard symbols and notation: for the pth value in a data set when the values are written in order, the sample lower and upper quartiles and the sample median, the sample mean and the standard deviation
- understand that data can have a pattern which may be represented graphically
- understand that the standard deviation and the interquartile range are measures of the dispersion in a data set
- understand that the median and the interquartile range are more resistant measures than are the mean and the standard deviation
- identify an overall 'feel' for data and the way it is distributed by constructing appropriate graphical displays.

Completed study

The learner has completed the following:

Section 1

Introducing data

Section 2

Data and questions

Section 3

Pie charts and bar charts

Section 4

Histograms and scatterplots

Section 5

Numerical summaries

Section 6

Conclusion